

[442]**SARDAR PATEL UNIVERSITY****BCA Semester-3 External Examination, October - 2025****US03CBCA54 [Old Course] || SYSTEM ANALYSIS AND DESIGN****Date :08-10-2025****Time : 02.00 p.m. – 05.00 p.m.****Total : 70 Marks****Q-1 Select the appropriate option for the following questions:****10**

- 1) The activity that makes possible the transformation of input to output is called _____.
a) output b) goal c) input d) process
- 2) The example of conceptual system is _____.
a) philosophy b) solar c) water d) computer
- 3) The third step of SDLC is _____.
a) Problem identification b) System evaluation
c) System requirement analysis d) System study
- 4) One of the example of onetime cost is _____.
a) Salary b) construction c) Equipment maintenance d) Supply
- 5) The processed data with specific purpose are called _____.
a) Data b) Information c) Picture d) System
- 6) _____ is a collection of data fundamentals to a system.
a) Data Base b) Data Input c) Data Output d) Data Entry
- 7) DFD stands for _____.
a) Data Flow Diagram b) Database Flow Diagram c) Data Flow Design d) None
- 8) The DFDs that shows "How the process is going on" is known as _____.
DFDs.
a) Physical b) Logical c) Physiological d) None
- 9) SSADM does not includes _____.
a) System Survey b) Structured Design c) Maintenance d) System Analysis
- 10) Arranging the data into some desired sequence is known as _____.
a) Data Recording b) Verification c) Sorting d) Data Transmission

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE)**20**

- 1) What is System? Give example of the system.
- 2) List Role of System Analyst.
- 3) Give examples of Natural and Artificial system.
- 4) List steps of SDLC.
- 5) Discuss technical feasibility.
- 6) Explain Survey in SSADM.
- 7) Define: Data and Data Collection
- 8) Explain term Validation.

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- 9) List basic steps for process of Data Capture.
- 10) What is Physical DFDs and Logical DFDs?
- 11) Write basic features of Prototypes.
- 12) Draw the Context level diagram for College Payroll System.

- Q-3** A List characteristics of System, explain any two in detail. **05**
 B Explain the elements of system. **05**

OR

- Q-3** A List the types of systems. Discuss any two of them in detail with example. **05**
 B Draw and explain the General Model of a system and model of Business System with diagrams. **05**

- Q-4** A Write a short note on System Analysis considering SDLC. **05**
 B What is system maintenance? Why is system maintenance some times both the longest and most expensive part of a system life cycle? **05**

OR

- Q-4** A Discuss advantages of SSADM. **05**
 B Discuss feasibility study with its need. **05**

- Q-5** A Write a note on Interview Techniques. **05**
 B Discuss method of Observation with its merits. **05**

OR

- Q-5** A Discuss design principles of Output. **05**
 B Discuss Questionnaires with its categories and use. **05**

- Q-6** Draw and discuss Symbols used in DFDs and explain thumb rules used to constructing DFDs. **10**

OR

- Q-6** Draw Prototype Development Model and discuss steps of Prototype Process in detail. **10**

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